

## Pattaraphon Kenny Wongchamcharoen

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| CONTACT INFORMATION        | 2117 Haste Street<br>Berkeley, CA 94704                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | +1 (510) 703 6076<br>pattaraphon.kenny@berkeley.edu |
| EDUCATION                  | <b>University of California, Berkeley</b><br>B.S. Industrial Engineering & Operations Research<br>Minors in Data Science and Mathematics<br>GPA: 3.98/4.00                                                                                                                                                                                                                                                                                                                                                                                                                                       | May 2027                                            |
| RESEARCH INTERESTS         | <b>Methods:</b> Stochastic Modeling and Queueing Theory; Optimization; Game Theory; Statistical and Empirical Methods; Applied Machine Learning<br><br><b>Applications:</b> Human-AI Operations; AI Reliability in Operations; Service and Transportation Systems; Platform Operations and Marketplace Design                                                                                                                                                                                                                                                                                    |                                                     |
| PUBLICATIONS & PROCEEDINGS | <div>[1] Do Large Language Models (LLMs) Understand Chronology?<br/><b>Wongchamcharoen, P. K.</b>, &amp; Glasserman, P.<br/><i>Proceedings of the 2026 AAAI Conference on Artificial Intelligence (AAAI-26)</i><br/>Oral – Student Abstract &amp; Poster Program (Top 11%)<br/>Poster – <i>AI4TS: AI for Time Series Analysis (AAAI-26 Workshop)</i><br/><a href="#">[proceedings]</a> <a href="#">[arXiv]</a> <a href="#">[code]</a></div>                                                                                                                                                      |                                                     |
| WORKING PAPERS & PREPRINTS | <div>[1] CentaurBench: Benchmarking LLM Capabilities on Augmenting vs. Automating Real-World Work Tasks<br/><b>Wongchamcharoen, P. K.</b>, Gulati, K., Fong, M. M., &amp; Nagaraj, A.<br/><i>NBER Working Paper #35663</i><br/><a href="#">[nber]</a> <a href="#">[arXiv]</a> <a href="#">[project dashboard]</a></div> <div>[2] On-Off Systems with Strategic Customers<br/>Sun, Y., <b>Wongchamcharoen, P. K.</b>, Liu, Z., &amp; Yan, C.<br/><i>Revise and Resubmit (R&amp;R) at Operations Research</i><br/><a href="#">[ssrn]</a></div>                                                     |                                                     |
| SELECTED RESEARCH PROJECTS | <div>[1] Data-Driven Evaluation of Board of Directors Effectiveness<br/><b>Wongchamcharoen, P. K.</b>, Stringer, C., Hwang, B., Liu, I., &amp; Li, K.<br/><i>Best Data Visualization, Berkeley CDSS Data Discovery Symposium 2025</i><br/><a href="#">[poster]</a></div> <div>[2] Mixed-Integer Linear Program for Options Pricing and Portfolio Optimization<br/>Bin Abdulla, Q. M., <b>Wongchamcharoen, P. K.</b>, Jamari, A., &amp; Lee, J.<br/><i>1st Runner-up, Wells Fargo &amp; IEOR Bay Area Decision Sciences Summit 2025</i><br/><a href="#">[poster]</a> <a href="#">[code]</a></div> |                                                     |
| RESEARCH CONTRIBUTIONS     | <div>[1] Calyber: A Ridesharing Game<br/>Shen, Y., Yan, C., &amp; Yan, J.<br/><i>INFORMS Transactions on Education</i>, 26(3):218–225, 2026<br/>Research assistant contribution; Runner-up, 2025 INFORMS Case Competition<br/><a href="#">[article]</a> <a href="#">[award]</a></div>                                                                                                                                                                                                                                                                                                            |                                                     |

## RESEARCH IN PROGRESS

- [1] Flex or Fast? Incentive-Compatible Demand Allocation in Destination-Mode Ride-Hailing Networks  
*In preparation*

## TALKS & PRESENTATIONS

Sun, Y., **Wongchamcharoen, P. K.**, Liu, Z., and Yan, C. (2026). “On-Off Systems with Strategic Customers.” Scheduled invited presentation at the *2026 INFORMS Annual Meeting, MSOM Invited Session* in San Francisco, CA.

**Wongchamcharoen, P. K.**, Gulati, K., Fong, M. M., and Nagaraj, A. (2026). “CentaurBench: Benchmarking LLM Capabilities on Augmenting vs. Automating Real-World Work Tasks.” Presentation at the *2026 Wharton Generative AI & Business Conference* in San Francisco, CA.

**Wongchamcharoen, P. K.** (2026). “AI Is Getting Smarter... Are We Cooked?: Research Perspectives on the Capabilities, Limits, and Future of AI.” Invited talk at the *Berkeley Engineering Summer Machine-Learning & AI Research Training (BeSMART) Program* in Berkeley, CA. [slides]

**Wongchamcharoen, P. K.** and Glasserman, P. (2026). “Do LLMs Understand Chronology?” Presentation at the *2026 IISE Annual Conference* in Arlington, TX.

**Wongchamcharoen, P. K.** and Glasserman, P. (2026). “The Temporal Blind Spots of AI: Implications for IEOR Applications.” Presentation at the *Berkeley IEOR Annual Advisory Board Meeting* in Berkeley, CA. [press]

**Wongchamcharoen, P. K.**, Gulati, K., Fong, M. M., and Nagaraj, A. (2026). “Centaur Benchmarking: Measuring the Value of LLM Assistance on Professional and Economically Valuable Tasks Using Simulation.” Presentation at the *Berkeley Haas Data Innovation and AI Lab (DIAL) Meeting* in Berkeley, CA.

**Wongchamcharoen, P. K.** and Glasserman, P. (2026). “Do LLMs Understand Chronology?” Oral presentation at the *40th Annual AAAI Conference on Artificial Intelligence* in Singapore. [poster] [talk]

**Wongchamcharoen, P. K.** and Glasserman, P. (2025). “Chrono-Logical? Assessing Temporal Reasoning in Large Language Models: An Empirical Lens.” Presentation at the *Columbia Business School Summer Research Seminar* in New York, NY.

Bin Abdulla, Q. M., **Wongchamcharoen, P. K.**, Jamari, A., and Lee, J. (2025). “Mixed-Integer Linear Program for Options Pricing and Portfolio Optimization.” Presentation at the *Berkeley IEOR Community & Alumni Achievement Ceremony* in Berkeley, CA.

## RESEARCH EXPERIENCE

**UC Berkeley Industrial Engineering & Operations Research**    Berkeley, CA  
*Research Assistant*  
*Advisor: Prof. Chiwei Yan*    Jan 2024 – Present

- Extended the on-off dynamic queueing model with strategic customers to multi-server systems; reformulated the nonconvex throughput problem as a linear program and proved theoretical results on throughput scaling, demand saturation, and optimal service allocation under exogenous on-off policies, including a closed-form linear-scaling threshold via an explicit dual certificate; contributed to the revision of *On-Off Systems with Strategic Customers* (R&R, *Operations Research*).
- Built and ran an end-to-end numerical experiment using Bay Area Rapid Transit (BART) Yellow Line origin–destination and GTFS data; estimated rider patience and recovered suppressed demand using a hierarchical Bayesian mixed-effects model.
- Optimized reusable dwell-time policies across 48 station-direction queues using sample-average approximation, projecting peak-period throughput gains of up to

24%; invited to present the findings in an MSOM invited session at the 2026 INFORMS Annual Meeting and contributed to securing a \$40,729 UC ITS research grant for passenger-centric BART schedule optimization.

- Designed and validated a discrete-event simulator with Poisson arrivals, strategic customer joining, exhaustive service, and counterfactual wait tracking; developed learning dynamics and algorithms for endogenous equilibrium computation; evaluated stochastic-to-fluid convergence and out-of-sample policy performance.
- Deployed *Calyber*, a 2025 INFORMS Case Competition runner-up, in a graduate supply-chain course; developed Python notebooks and teaching materials; guided teams in designing pricing and matching policies using a dataset of more than 10,000 Chicago rider arrivals, supporting the case's subsequent publication in *INFORMS Transactions on Education*.
- Built an AI-agent competitor and evaluation workflow to benchmark agentic policies against student teams on profit, conversion, matching, and cost efficiency.

**UC Berkeley Haas School of Business**

Berkeley, CA

*Research Assistant, Data Innovation and AI Lab (DIAL)*

*Advisor: Prof. Abhishek Nagaraj*

Jan 2024 – Present

- Led a four-person research team to conceptualize and develop *CentaurBench*, a novel benchmark distinguishing an LLM's ability to automate professional work from its ability to augment another agent.
- Designed and ran experiments across seven professional tasks and ten frontier models, finding that automation and augmentation rankings diverge; NBER Working Paper #35663; presented at the 2026 Wharton Generative AI & Business Conference.
- Architected a multi-agent workflow and engineered a multi-provider Python API pipeline spanning OpenAI, Anthropic, Google, DeepSeek, Together AI, and Expected Parrot, with asynchronous execution, token controls, retries, timeouts, and automated detection of truncated or malformed outputs.
- Built a reproducible LLM-as-judge harness using anonymized round-robin comparisons, randomized response ordering, micro-rubrics, four-model judge panels, and *leave-family-out* masking; validated 15,000+ scored comparisons using rank correlations, inter-judge agreement, Krippendorff's alpha, and sensitivity analysis.
- Created a dashboard and reproducibility repository auditing outputs, rankings, rationales, and replication diagnostics.
- Conducted foundational research on generative AI and online platforms: built logistic regression and random forest classifiers distinguishing human versus AI OpenStreetMap contributions; reviewed literature on copyrighted training data and LLM-as-judge evaluation.

**Columbia Business School, Decisions, Risk, and Operations** New York, NY

*Summer Research Fellow*

*Advisor: Prof. Paul Glasserman*

Jun 2025 – Aug 2025

- Selected as 1 of 15 interns from more than 1,000 applicants for Columbia Business School's Summer Research Internship Program to investigate whether large language models (LLMs) understand chronology.
- Built and deployed a reproducible Python pipeline using GPT-4.1, o3, Claude Sonnet 3.7, and other frontier LLM APIs to simulate event-ordering tasks and evaluate chronological reasoning at scale across increasingly complex task families: basic sorting, conditional sorting, and anachronism detection.

- Developed a statistical evaluation framework comparing reasoning models using Spearman rank correlation, Kendall rank correlation, and complementary rank-based diagnostics across model families and task configurations.
- Published the findings in the AAAI-26 proceedings and presented them orally in the Student Abstract Program (top 11%) and as a poster at the AI4TS: AI for Time Series Analysis Workshop; invited to present at the 2026 IISE Annual Conference and Berkeley IEOR Annual Advisory Board Meeting; cited in OpenAI’s 2026 paper, “Scaling social science research” [*working paper*].

**UC Berkeley School of Law** Berkeley, CA  
*Student Researcher*  
*Advisor: Prof. Frank Partnoy* Jan 2025 – May 2025

- Scraped corporate board skills matrices from SEC DEF 14A filings and applied unsupervised learning techniques, including t-SNE clustering, predictive modeling, and data visualization to develop data-driven measures of board effectiveness.
- Presented the findings at the Berkeley CDSS Data Discovery Symposium 2025, receiving the Best Poster: Data Visualization award.

**UC Berkeley Haas School of Business** Berkeley, CA  
*Research Assistant*  
*Advisor: Prof. David Evan Harris* Aug 2023 – Jan 2024

- Conducted literature reviews on ethical and governance questions surrounding artificial intelligence and its organizational use; collected and organized source articles and citations using Zotero; presented findings in weekly lab meetings.
- Supported manuscript preparation by proofreading drafts, checking citations, and helping address reviewer comments and rebuttal materials.

PROFESSIONAL **Beats by Dre** Culver City, CA  
 EXPERIENCE *Data Analytics Extern* Jun–Aug 2024

- Conducted exploratory data analysis and sentiment analysis on more than 1,000 Amazon reviews of Beats Pill speakers using TextBlob, NLTK, and Scrapy.
- Utilized advanced NLP techniques and Google’s Gemini API to extract, summarize, and analyze customer feedback; designed and presented Tableau dashboards highlighting customer sentiment and purchase trends.

**Xcube Capital** Singapore  
*Product Development & Analytics Intern* May–Aug 2024

- Scraped and analyzed job postings from JobsTimes using BeautifulSoup to identify emerging career trends and skill gaps; built machine-learning models forecasting HR-technology job demand with more than 80% accuracy.
- Automated internal workflows using SQL queries to optimize the company’s CRM database; integrated email scheduling via SMTP and Python’s `schedule` library, reducing manual processing time by 90%.
- Conceptualized a proof of concept for a proprietary asset class—Corporate Venture Portfolio Equities (*CVPE*)—targeting investments in behavioral-AI startups.

**MENTOR Global Consultants**, Technical Product Mgmt. Intern Jan–Apr 2024  
**GUESS, Inc.**, Strategy Contract Consultant Aug–Dec 2023

HONORS AND **Tau Beta Pi Scholarship** (Tau Beta Pi Association; \$2,000) 2026  
 AWARDS

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|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
|                             | Bonder Foundation Scholarship (INFORMS; \$400)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2026                    |
|                             | Dean's Lists (UC Berkeley College of Engineering)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2023–2026               |
|                             | CBS Research Internship Grant (Columbia Business School; \$7,500)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2025                    |
|                             | Best Poster: Data Visualization Award (UC Berkeley CDSS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2025                    |
|                             | 1st Runner-up, Bay Area Decision Sciences Summit (Wells Fargo; \$600)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025                    |
|                             | Cal Alumni Leadership Award Finalist (Cal Alumni Association)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2024                    |
|                             | College of Engineering Mentorship Grant (UC Berkeley; \$850)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2024                    |
|                             | ASEAN Scholarship (Singapore Ministry of Education; full 4-year award)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2018                    |
| TEACHING<br>EXPERIENCE      | <b>Tutor &amp; Grader</b> , University of California, Berkeley                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Fall 2024–Present       |
|                             | <ul style="list-style-type: none"> <li>• IEOR 145: Fundamentals of Revenue Management (Prof. Rajan Udwani)</li> <li>• IEOR 142A: Machine Learning &amp; Data Analytics I (Prof. Phillip Kerger)</li> <li>• IEOR 172: Probability &amp; Risk Analysis for Engineers (Prof. Daniel Pirutinsky)</li> <li>• IEOR 120: Principles of Engineering Economics (Prof. Daniel Pirutinsky)</li> <li>• CS 88: Computational Structures in Data Science (Prof. John DeNero)</li> <li>• MATH 1A, 1B, 53, and 54; DATA 8 and 100; CS 61A; IEOR 160 and 162 for Berkeley's Center for Access to Engineering Excellence (CAEE)</li> </ul> |                         |
|                             | <b>Primary Lecturer</b> , Berkeley Engineering GLOBE Center                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Fall 2025–Present       |
|                             | <ul style="list-style-type: none"> <li>• Developed and taught <i>Foundations of Machine Learning</i>, an introductory machine-learning course for underserved Bay Area high-school students (Program Advisor: Prof. Phillip Kerger). [<a href="#">course website</a>]</li> </ul>                                                                                                                                                                                                                                                                                                                                         |                         |
|                             | <b>Teaching Assistant</b> , University of California, Berkeley                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Spring–Summer 2026      |
|                             | <ul style="list-style-type: none"> <li>• IEOR C253: Supply Chain &amp; Logistics Management (Prof. Chiwei Yan)</li> <li>• Berkeley Summer Machine Learning &amp; AI Research Training Program (BeSMART); mentored student machine learning research projects and presentations, created assignments, and guest lectured on coding and agentic AI (Prof. Phillip Kerger)</li> </ul>                                                                                                                                                                                                                                       |                         |
| SERVICE AND<br>LEADERSHIP   | <b>Institute of Industrial &amp; Systems Engineers at Berkeley</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                         |
|                             | Senior Advisor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Fall 2025 – Present     |
|                             | Director of Programs and Events                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Fall 2023 – Spring 2024 |
|                             | <b>Singapore Students Association at Berkeley</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                         |
|                             | Vice President of External Relations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Spring 2024 – Fall 2024 |
| PROFESSIONAL<br>MEMBERSHIPS | Institute for Operations Research and the Management Sciences (INFORMS; Student Member); Institute of Industrial & Systems Engineers (IISE; Student Member); Tau Beta Pi Engineering Honor Society (Member)                                                                                                                                                                                                                                                                                                                                                                                                              |                         |
| TECHNICAL<br>SKILLS         | Python (NumPy, Pandas, Matplotlib, scikit-learn, Pyomo, TensorFlow); Java; SQL; Apache Spark; LLM APIs (OpenAI, Anthropic, Google Gemini); HuggingFace; L <sup>A</sup> T <sub>E</sub> X                                                                                                                                                                                                                                                                                                                                                                                                                                  |                         |
| LANGUAGES                   | English (Fluent); Thai (Native); Mandarin Chinese (Elementary)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                         |